

Acquisition phase-specific contribution of climbing fiber transmission to cerebellum-dependent motor memory

Reviewed Preprint

v2 • April 9, 2025

Revised by authors

Reviewed Preprint

v1 • May 23, 2024

Jewoo Seo, Seung Ha Kim, Jaegeon Lee, Min Seok Kim, Yong-Seok Lee, Sang Jeong Kim 

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea • Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea • Memory Network Medical Research Center, Neuroscience Research Institute, Wide River Institute of Immunology, Seoul National University College of Medicine, Seoul, Republic of Korea

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

This study presents potentially **valuable** insights into the role of climbing fibers in cerebellar learning. The main claim is that climbing fiber activity is necessary for optokinetic reflex adaptation, but is dispensable for its long-term consolidation. There is evidence to support the first part of this claim, though it requires a clearer demonstration of the penetrance and selectivity of the manipulation. However, support for the latter part of the claim is **incomplete** owing to methodological concerns, including the robustness of the CF marking and manipulation approach and the unclear efficacy of longer-duration climbing fiber activity suppression.

<https://doi.org/10.7554/eLife.95838.2.sa3>

Abstract

Climbing fiber (CF) transmission from the inferior olivary triggers complex spikes (Cs) in Purkinje cells (PCs) driven by a burst of calcium spikes. In the context of motor learning, especially the compensatory optic response, CF transmission serves as instructive signals selectively conveyed to PCs. While the significance of CF input in motor memory formation is widely acknowledged, a comprehensive understanding of its distinct contribution across different temporal windows, spanning from the initial learning phase to the retrieval period, remains incomplete. Therefore, we aimed to investigate the necessity of CF-induced instructive signals in motor learning by assessing their roles in memory acquisition, consolidation, and retrieval. We employed optogenetics to selectively inhibit CF transmission during targeted time windows. Consequently, the absence of CF-induced instructive signals during motor learning impairs memory acquisition. However, when these signals were suppressed during the consolidation and retrieval period, there was neither a loss of long-term memory nor prevention of memory retrieval. Our results highlight that CF transmission plays a specialized and critical role primarily in memory acquisition, rather than subsequent processes.

Introduction

The cerebellar architecture, characterized by multiple synapses connecting diverse neuron types, enables precise adjustments within intricate neural networks (Boyden et al., 2004). Within this framework, the climbing fiber (CF) transmission derived from the inferior olive (IO) projecting to the cerebellar cortex is generally recognized as the primary pathway for relaying instructive signals to correct errors and enhance motor performance, including oculomotor behaviors (Albus, 1971; Frens et al., 2001; Ito, 1972; Marr, 1969).

A prominent example of cerebellum-dependent motor learning is the optokinetic reflex (OKR), a compensatory optic response evoked by an unstable visual field (Cahill & Nathans, 2008; Schweigart et al., 1997). In OKR, adaptation to stabilize retinal images amid mismatched visual input requires the engagement of multiple neural substrates. For instance, complex spikes (Cs) in floccular Purkinje cells (PCs) driven by CF input, emerge in response to unpredicted motor performance errors (Clopach et al., 2014; Eccles et al., 1966; Goossens et al., 2004; Palmer et al., 2010). Rotational visual stimulation induces Cs discharge in floccular PCs, reflecting CF-induced Cs generation in response to error signals generated from errors in oculomotor performance (Frens et al., 2001; Graf et al., 1988). Therefore, this fine-tuning of Cs in PCs is expected to regulate learning by modulating error-correcting adaptations (Yang & Lisberger, 2014).

Previous studies highlight a potential duality in CF involvement in cerebellar motor learning. For example, one study using a modified behavioral paradigm to suppress Cs suggested that simple spikes (Ss) alone may suffice for motor learning (Ke et al., 2009). In contrast, pharmacological lesioning and optogenetic inactivation of the IO impair oculomotor learning (Pham et al., 2020) and prevent eye blink conditioning (Kim et al., 2020; Medina et al., 2002; Welsh & Harvey, 1998). While CF's role in acquisition is well-supported by existing literature, its involvement in downstream memory processes like consolidation and retrieval remains contentious. This ambiguity stems from methodological limitations in previous studies, which have lacked the spatial and temporal precision to isolate CF function at distinct learning stages. Addressing this knowledge gap is essential to clarify whether CF input is uniformly critical or phase-specific in motor learning and memory.

To address this gap, we investigated the necessity of CF-induced instructive signals in motor learning by assessing their contributions across memory acquisition, consolidation, and retrieval phases. Using optogenetics, we selectively inhibited CF signals in the flocculus, a lobe-like cerebellar structure primarily responsible for ocular reflex, during targeted learning stages. Our findings provide compelling evidence for a phase-specific role of CF transmission in motor learning. This indicates that CF input is essential primarily during acquisition, with minimal involvement in consolidation and retrieval.

Results

Optogenetic inhibition of climbing fiber transmission derived from the inferior olive

To specifically manipulate CF signals in the cerebellar cortex, generally known to originate from the IO, we injected AAV1-CaMKII α -eNpHR 3.0-EYFP or AAV1-CaMKII α -EGFP into the IO of wild-type mice (Figure 1A, B). This approach allowed visualization of IO neuron expression and CF terminals in the brainstem nucleus and flocculus. Importantly, the selectively segregated

expression in these regions enabled manipulation of CF terminals without disrupting IO somas, which, if damaged, can critically impair motor performance and ocular reflexes due to the IO's role as a prominent center of sensory integration (Pham et al., 2020 [↗](#); Shaikh et al., 2017 [↗](#); Van Der Giessen et al., 2008 [↗](#)).

To confirm the effectiveness of optogenetic inhibition of climbing fiber (CF) transmission, we performed whole-cell patch-clamp recordings of Purkinje cells (PCs) in acute slices of cerebellar vermis lobules 4–5. A 593 nm yellow laser was used to activate halorhodopsin (NpHR). PCs were selected based on optical fluorescence expression (**Figures 1C** [↗](#) and **1E** [↗](#)), and a stimulation electrode was positioned in the granule cell layer near the recorded PC. The stimulation intensity was carefully calibrated to evoke minimal CF excitatory postsynaptic currents (EPSCs). Laser illumination (3–5 mW) during stimulation resulted in a significant reduction in EPSC amplitude, indicating robust inhibition of CF transmission. Notably, this inhibition was consistent, with 9 out of 9 cells in the NpHR group exhibiting reliable suppression without failures. Importantly, laser illumination did not affect CF EPSCs in the GFP group (**Figure 1D** [↗](#)). Under the current-clamp mode, CF stimulation-induced spikelets were also completely abolished in the NpHR group but remained unaffected in the GFP group (**Figure 1F** [↗](#)). These findings establish that optogenetic inhibition effectively suppresses CF-PC synaptic transmission in vitro.

To extend our findings in vivo, we recorded from PCs in awake, head-fixed mice during 593 nm optogenetic stimulation. AAV1-CaMKIIα-eNpHR 3.0-EYFP or AAV1-CaMKIIα-EGFP was injected into the IO followed by the creation of a cranial window above the cerebellar vermis three weeks later (**Figure 2A** [↗](#)). Due to technical constraints preventing direct access to the flocculus for in vivo recordings, we targeted cerebellar lobule IV/V. PCs were identified based on the detection of Cs (**Figure 2B** [↗](#)).

Optogenetic inhibition selectively suppressed Cs firing rates in the NpHR group without affecting Ss rates, whereas no changes were observed in the GFP control group (**Figures 2C** [↗](#) and **2D** [↗](#)). Recordings from the same neurons during both laser-off and laser-on conditions demonstrated consistent and robust suppression of Cs activity throughout the 40-minute optostimulation period. Although continuous recordings over several hours were not feasible, the stability and sustained suppression observed at 40 minutes strongly suggest that the manipulation would remain effective during the extended durations required for behavioral experiments. These findings confirm that optogenetic inhibition of NpHR-expressing CFs selectively suppresses Cs activity in awake mice without altering Ss activity. Furthermore, they validate the feasibility and specificity of using optogenetic interventions to block CF transmission both in vitro and in vivo, reinforcing the reliability of this approach for studying CF contributions to cerebellar function and behavior.

Inhibition of climbing fiber transmission during memory acquisition

An evaluation of CF's direct contribution in a region-specific manner has not been performed. Thus, after confirming the viability of optogenetic inhibition of CF, we investigated whether the elimination of CF transmission in the cerebellar flocculus affects cerebellar motor learning. When images on the retina are unstable due to mismatched visual inputs, CF signals are conventionally perceived to be responsible for sending error signals to the cerebellar cortex to improve ocular reflexes (Bloedel & Bracha, 1998 [↗](#); Frens et al., 2001 [↗](#); Zang & De Schutter, 2019 [↗](#)). Using the same mouse model shown in **Figure 1** [↗](#), we surgically implanted an optical fiber in flocculus to suppress CF transmission in vivo during OKR (**Figure 3A** [↗](#)). Head-fixed mice were placed in a rotating drum painted with a black striped pattern to provide visual input, and its rotating motion elicited an unstable retinal image. Hence, in the beginning, the trace of the mouse's eye movement represented in a sine curve was only partially overlapping with the trace of the drum's kinetics, but as the mouse underwent more sessions of OKR learning, the trace of eye movement became

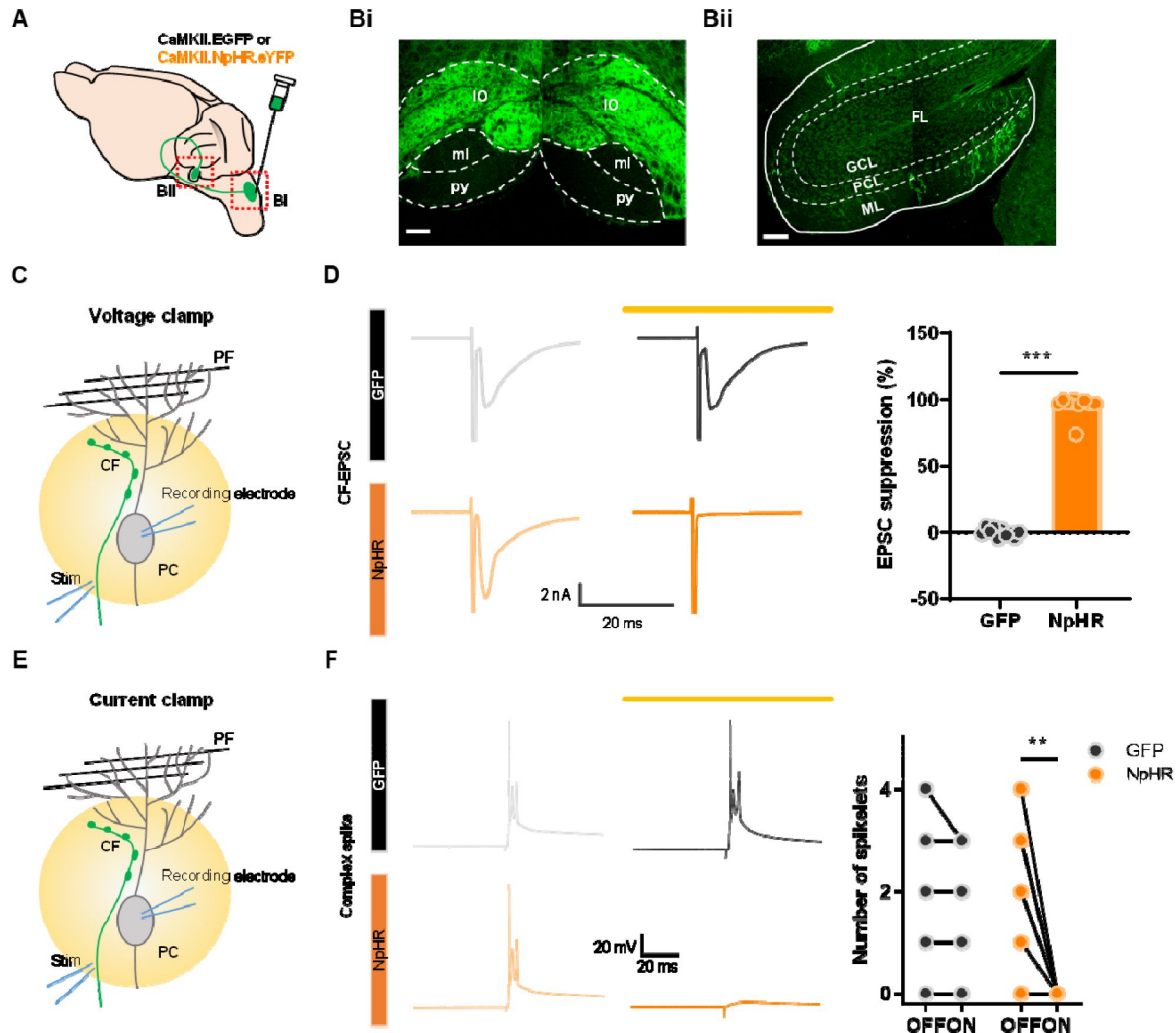


Figure 1.

Optogenetic inhibition of climbing fiber transmission derived from the inferior olive.

- A. Schematic diagram showing virus injection for optogenetic manipulation of CF transmission.
- B. Bi. A virus is expressed specifically only in the IO region (IO, inferior olive; py, pyramidal tract; ml, medial lemniscus). Scale bar, 100 μ m. Bii. Virus expression along CF in the cerebellar flocculus region. (GCL, granule cell layer; PCL, Purkinje cell layer; ML, molecular layer; FL, flocculus). Scale bar, 100 μ m.
- C. A scheme of optogenetic suppression using a yellow laser (593 nm, 3–5 mW) while recording CF EPSCs under voltage clamp mode.
- D. Left, representative traces of CF EPSCs of the GFP (top) and NpHR group (bottom) with (dark line) and without (light line) opto-stimulation. Right, Quantitative analysis of suppression of CF EPSCs (GFP: n = 10 cells/2 mice, NpHR: n = 9 cells/5 mice, *** p < 0.001, Unpaired t-test).
- E. A scheme of optogenetic suppression while recording complex spikes under current clamp mode.
- F. Representative traces of complex spike of the GFP (top) and NpHR group (bottom) with (dark line) and without (light line) opto-stimulation. Quantitative analysis of number of complex spike spikelets (GFP: n = 8 cells/1 mouse, NpHR: n = 8 cells/5 mice, ** p = 0.002; Paired t-test).

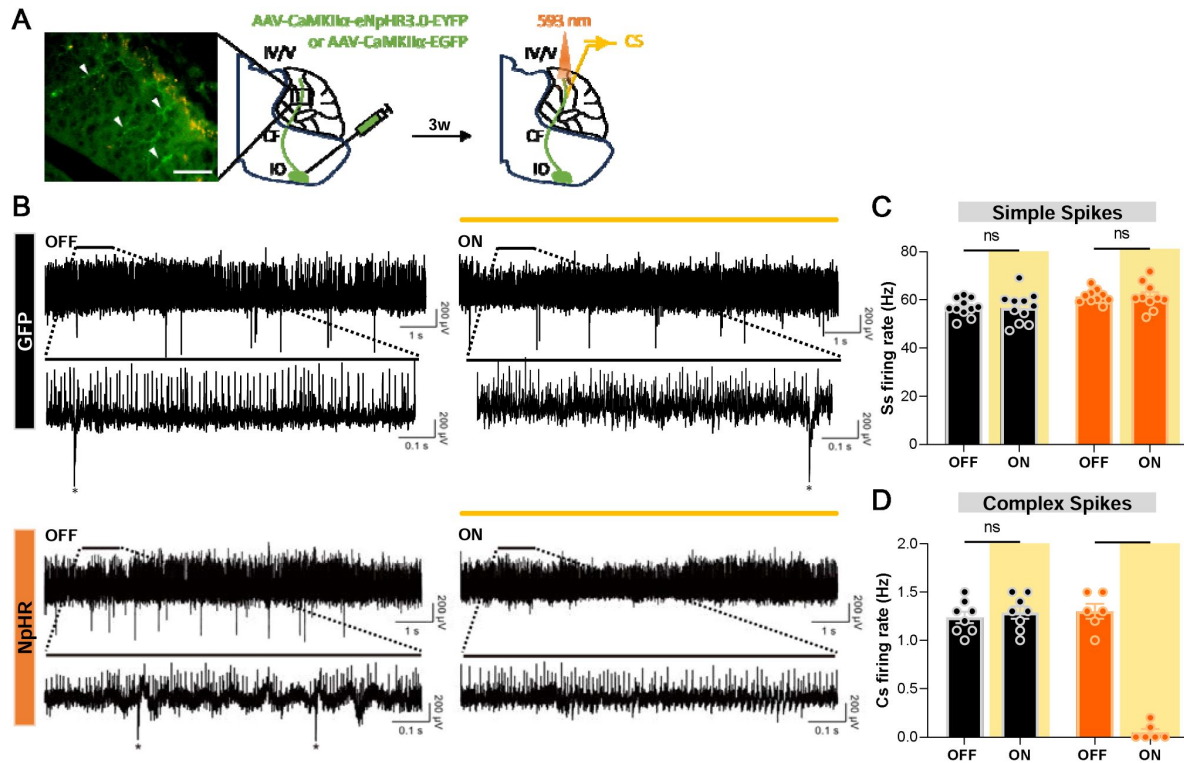


Figure 2.

Optogenetic inhibition of climbing fiber transmission blocked Cs firing but not Ss firing in vivo

A. Schematic diagram of virus injection into IO and in vivo recording of CF transmission. Arrow heads indicate NpHR-expressing CF terminals in the molecular layer of the cerebellar cortex. Scale bar, 50 μ m.

B. Representative electrophysiological recording traces of PC firing from the GFP and NpHR groups, without (OFF) and with (ON) yellow (593 nm) opto-stimulation. The upper plots show 10-second traces without and with opto-stimulation, and the lower plots display a 1-second segment from the respective upper traces. Asterisks denote Cs.

C. Comparison of Ss firing rates during optogenetic inhibition. For the GFP group, firing rates with laser off ($n = 11$ cells from 4 mice) versus on ($n = 11$ cells from 4 mice) were not significantly different (Mann-Whitney test, $P = 0.8594$). Similarly, for the NpHR group, no significant difference was observed between OFF ($n = 10$ cells from 6 mice) and ON ($n = 10$ cells from 6 mice) conditions (Mann-Whitney test, $p > 0.9999$).

D. Comparison of Cs firing rates during optogenetic inhibition. In the GFP group, no significant difference was observed between OFF ($n = 8$ cells from 4 mice) and ON ($n = 8$ cells from 4 mice) conditions (Mann-Whitney test, $p = 0.4563$). In the NpHR group, Cs firing rates were significantly reduced during opto-stimulation (OFF, $n = 6$ cells from 6 mice; ON, $n = 6$ cells from 6 mice; Mann-Whitney test, $** p = 0.0022$). Of note, 6 out of 8 recorded cells were responsive to optogenetic suppression.

almost identical (**Figure 3Bi** [↗](#)). Notably, the group with inhibited CF transmission exhibited only slight changes or no improved trace of eye movements (**Figure 3Bii** [↗](#)). Overall, the results demonstrated that CF transmission is required for motor memory acquisition.

This indicates that CF transmission is essential for cerebellar motor learning by presenting two contrasting learning curves. The control group exhibited an increasing trend in the learning curve, whereas the NpHR group's learning curve was almost flat, which reflects no change of gain (**Figure 3C** [↗](#)). The gain value through the learning sessions was distinctly increased in the control group, whereas the NpHR group showed no significant change in gain (**Figure 3D** [↗](#)). Furthermore, from 0 to 50 min period of learning, the control group demonstrated almost a doubling in gain, whereas the NpHR group showed little or almost no change of gain (**Figure 3E** [↗](#)). These results indicate that CF transmission is required for the proper acquisition of motor memory.

Inhibition of climbing fiber transmission during memory consolidation and retrieval phase

When memories are initially acquired, they are generally unstable; hence, it requires some time to shape a stabilized long-term memory (Attwell et al., 2002 [↗](#)). Therefore, memories undergo multiple processes that allow distinct shifts from short-term to long-term memory, leading to the construction of adapted behavior (Shutoh et al., 2006 [↗](#)). Having confirmed the active involvement of CF transmission during memory acquisition, we sought to understand if its role is limited to the acquisition period or extends to memory transfer during the consolidation phase for long-term storage. We implemented identical surgical and behavioral procedures as previously conducted. However, instead of optogenetically inhibiting the transmission during memory acquisition, we suppressed CF activity in two distinct optogenetic manipulation schemes: one targeting short-term inhibition (30 min) and the other for long-term inhibition (6 hr) (**Figures 4A** [↗](#) and **4D** [↗](#)). This approach was based on findings that long-term memories associated with cerebellum-dependent behaviors are vulnerable to disruption within specific time windows of consolidation, spanning from minutes to hours (Cooke et al., 2004 [↗](#); Titley et al., 2007 [↗](#)). Learning curves confirmed the successful gain enhancement, validating the integrity of memory acquisition (**Figures 4B** [↗](#) and **4E** [↗](#)). Once the gain was robustly enhanced, optogenetic inhibition of CF transmission during consolidation did not yield any change in the maintenance of memory, irrespective of the duration of manipulation. Consequently, long-term memory was well-established in both NpHR and GFP groups (**Figures 4C** [↗](#) and **4F** [↗](#)). In addition, we verified whether CF signaling is required during the retrieval phase (**Figure 4G** [↗](#)). When mice, fully trained in the optokinetic reflex, displayed increased gain, disrupting CF signaling did not affect the retrieval of motor memory (**Figure 4H** [↗](#)). This divergence underscores a specialized role for CF transmission in the early stages of motor learning, with subsequent phases relying on distinct neural mechanisms. The results highlight phase-specific contributions of CF-driven error signaling, aligning with the hypothesis that memory consolidation and retrieval depend on downstream circuits independent of CF activity.

Discussion

Motor learning is a finely tuned process, yet its underlying cellular mechanisms operate with remarkable precision, adapting selectively to task demands. This adaptability relies on CF activity, originating from the inferior olive, which instructs cerebellar circuits to reduce motor errors and enhance performance. CF-driven Cs in PCs have long been associated with motor learning, but how CF signals vary across learning stages remains less understood. Our study uncovers a distinct role for CF signaling that challenges traditional views: CF input is essential for driving motor adaptation during memory acquisition but surprisingly dispensable during consolidation and

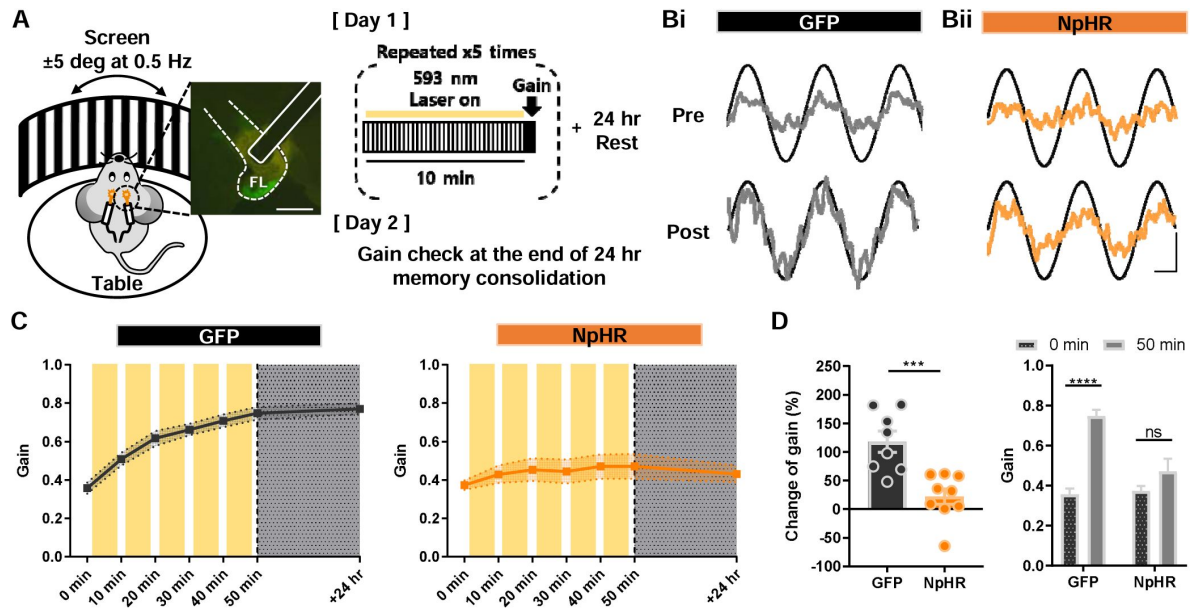


Figure 3.

Inhibition of climbing fiber transmission during memory acquisition of OKR.

A. Illustration of the optokinetic reflex behavioral test (left) and experimental scheme (right). For training, visual stimulation was given by constant rotation of the screen, while the table, where a mouse's head is fixed, remained stationary. On day 1, a mouse was trained for a total of 50 minutes, and on day 2, an amount of sustained memory was validated via gain check. Scale bar, 100 μ m

B. Representative traces of the screen (black sinusoidal curve) and eye movements (grey sinusoidal curve) divided into the GFP (i) and NpHR (ii) groups. Top traces (pre) were acquired before learning, and bottom traces (post) were obtained after 50 min of learning. The vertical scale bar represents 10 degrees per second and the horizontal scale bar represents 0.5 seconds.

C. The learning curve from 0 min to 50 min, and +24 hr period. Change of gain is indicated at each time point (+10 min increment). Yellow boxes indicate opto-stimulation (12.5 mW for 10 min/session, total of 5 sessions). The left graph represents the GFP (n = 8) and the right graph indicates the NpHR (n = 9) group.

D. Comparison of gain changes between the GFP and NpHR groups (GFP, n = 8 mice; NpHR, n = 9 mice). Percentages of gain increment from 0 min to 50 min were calculated. The change of gain of the GFP group was significantly larger than that of the NpHR group (left; Unpaired t-test, $p = 0.0007$). The gain of the GFP group was significantly increased from 0 min to 50 min (right; Paired t-test, **** $p < 0.0001$), while the NpHR group showed no significant improvement of gain from 0 min to 50 min (right; Paired t-test, $p = 0.1046$).

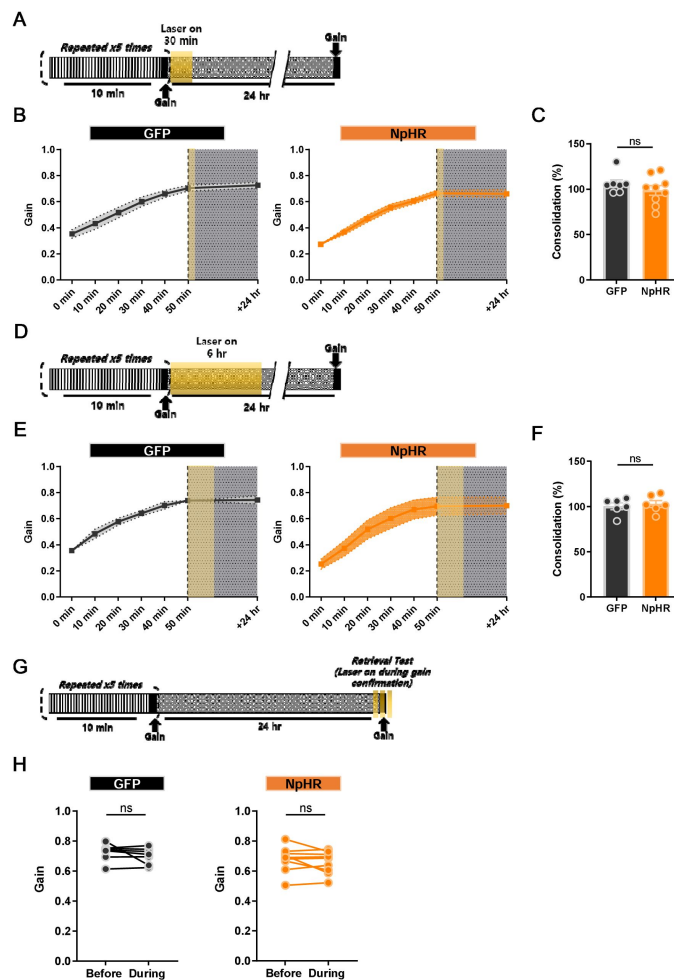


Figure 4.

Inhibition of climbing fiber transmission during memory consolidation and retrieval phase.

A. Illustration of the experimental scheme. 10 min of optokinetic reflex training was repeated 5 times, and then opto-stimulation (12.5 mW) was given at 0 min post-learning for 30 min.

B. The learning curve from 0 min to 50 min, and +24 hr period. Change of gain is indicated at each time point (+10 min increment). A yellow box indicates opto-stimulation at 0 min post-learning. The left graph represents the GFP and the right graph indicates the NpHR group.

C. Comparison of consolidation percentage between the GFP and NpHR groups (GFP group, n = 7 mice; NpHR group, n = 9 mice). The percentage of sustained memory after 24 hr was calculated. No difference was observed between the groups (p = 0.3173; Unpaired t-test).

D. Illustration of the experimental scheme. 10 min of optokinetic reflex training was repeated 5 times, and then opto-stimulation (12.5 mW) was given at 0 min post-learning for 6 hr.

E. The learning curve from 0 min to 50 min, and +24 hr period. Change of gain is indicated at each time point (+10 min increment). A yellow box indicates opto-stimulation at 0 min post-learning. The left graph represents the GFP group, and the right graph indicates the NpHR group.

F. Comparison of gain changes between GFP and NpHR groups (GFP group, n = 6 mice; NpHR group, n = 6 mice). Percentages of gain increment from 0 min to 50 min were calculated. No difference was observed between the groups (Unpaired t-test, p = 0.6405).

G. Illustration of the experimental scheme. Following proper motor learning on day 1, the remaining gain value was checked on day 2, after testing 1st gain retrieval without opto-stimulation, 2nd gain retrieval was tested with opto-stimulation to validate the effect of CF inhibition on gain retrieval (opto-stimulation was given 3 times for 24 seconds each at 12.5 mW).

H. Gain retrievals before and during opto-stimulation were compared. The left graph represents the GFP group, and the right shows the NpHR group (GFP group, n = 6 mice; NpHR group, n = 8 mice). There was no significant variation between the groups (NpHR group, p = 0.2439; GFP group, p = 0.3180; Paired t-test).

retrieval. This phase-specific pattern of CF transmission in cerebellum-dependent learning indicates that its role is not uniform over time, as previously thought, but instead selectively tunes cerebellar circuits to optimize adaptation.

Through optogenetic inhibition of CF transmission, we found that CF activity is critical for enhancing motor gain during initial memory acquisition, where substantial adjustments and error corrections are required. In contrast, CF input played a minimal role during memory consolidation and retrieval, suggesting that other mechanisms may take precedence as motor memories stabilize and transfer to downstream circuits. This discovery underscores a refined model of CF function: CFs initiate learning adjustments but are not integral to later stages of memory storage or retrieval, reshaping our understanding of the cerebellum's contribution to motor memory over time.

Motor memories acquired through cerebellar-dependent learning are believed to transfer to the vestibular nuclei for long-term storage (Jang et al., 2020 [↗](#); Seo et al., 2024 [↗](#); Shutoh et al., 2006 [↗](#)). By applying optogenetic inhibition of CF transmission at various learning stages, we differentiated the phase-specific role of CF signaling. CF activity proved indispensable for motor adaptation (i.e., gain increase) during acquisition, yet was unnecessary during consolidation and retrieval, suggesting CF input's primary role is in initial memory formation rather than in later memory processing.

Our findings also indicate that CF-dependent and CF-independent mechanisms work in a context-sensitive manner during cerebellar learning, consistent with established insights from the vestibulo-ocular reflex (VOR) and OKR systems. Specifically, CF signaling appears essential in tasks involving substantial, persistent errors that require precise error correction. For instance, in the VOR, gain increases depend on CF-driven complex spikes, which provide adaptive signals in response to large error inputs (Ke et al., 2009 [↗](#)). Conversely, CF-independent processes can support VOR gain decreases, where smaller or more gradual adjustments are sufficient. This selective role of CF signaling reflects a broader functional adaptation within cerebellar circuits, allowing different pathways to be recruited according to motor learning demands. Our OKR findings align with this, highlighting CF-dependent pathways in tasks with significant error correction during acquisition, while CF-independent mechanisms likely support incremental, lower-error adjustments, facilitating efficient motor adaptation without high-frequency CF input. This division of labor between CF-dependent and CF-independent processes underscores the cerebellum's adaptive plasticity, where CF input selectively engages based on the scale and persistence of motor errors. By dynamically alternating between these mechanisms, the cerebellum achieves a balance between precision and flexibility, optimizing motor adaptation across various behavioral contexts. This interplay provides a versatile framework for motor learning across cerebellar-dependent tasks, extending from VOR to OKR and likely other sensorimotor adaptations.

The process of systems consolidation, facilitated by the medial vestibular nuclei (MVN) and downstream circuitry of floccular PCs, is essential for long-term memory storage (Shutoh et al., 2006 [↗](#)). Given that CF activity modulates floccular PC firing, we hypothesized that CF signaling might indirectly influence MVN activity during memory consolidation and retrieval. However, contrary to this expectation, our results revealed no substantial effect of CF transmission on long-term memory formation or recall, suggesting that CF signaling does not directly impact MVN function. Instead, previous studies suggest that PC intrinsic plasticity supports memory consolidation (Seo et al., 2024 [↗](#)), while granule cell activity contributes to memory retrieval (Wada et al., 2014 [↗](#)). These findings reinforce the notion that CF signaling primarily facilitates the acquisition phase of motor memory, while simple spiking in PCs may encode timing and frequency information crucial for stabilizing memory traces in downstream circuits. SCs could either complement or function independently of CF transmission, especially during phases when error correction demands are minimal. Mechanistically, suppressing CF signaling likely mimicked key

aspects of long-term depression (LTD) at CF-PC synapses, such as reducing the amplitude of Cs (Hansel & Linden, 2000 [DOI](#)). Under normal conditions, CF input promotes parallel fiber-PC (PF-PC) LTD by driving calcium influx during error correction, such as in response to retinal slip (Coessmans et al., 2004 [DOI](#)). CF-LTD modifies Cs waveforms and attenuates CF-evoked dendritic calcium transients (Weber et al., 2003 [DOI](#)). In our study, optogenetic CF suppression likely simulated the functional consequences of CF-LTD, which may have impaired PF-PC LTD and subsequently hindered gain enhancement during learning (Han et al., 2007 [DOI](#)).

Our findings underscore a phase-specific role for climbing fiber (CF) input, demonstrating that CF activity is critical for motor memory acquisition but less necessary for consolidation and retrieval. Using optogenetic inhibition, we achieved precise temporal control over CF activity within distinct learning phases, enabling targeted manipulation of CF signaling. This differs from prior studies that employed pharmacological or lesion methods, which affect broader cerebellar pathways and may implicate CF signaling across multiple memory stages. The task demands inherent to different learning phases may explain these observed differences. Tasks requiring high-error correction, such as those encountered during the active learning phase, likely depend on CF-driven signals for precise error correction during acquisition. In contrast, CF-independent mechanisms may dominate later phases where such adjustments are less critical. This context-sensitive role highlights the adaptability of CF signaling in tuning cerebellar circuits to meet the specific demands of each learning phase. Moreover, cerebellar-dependent memory consolidation exhibits biphasic temporal dynamics: an initial rapid phase of consolidation followed by prolonged stabilization. Our study specifically focused on the early consolidation phase, emphasizing cerebellar plasticity mechanisms that are critical for encoding and stabilizing motor memory (Cooke et al., 2004 [DOI](#); Seo et al., 2024 [DOI](#); Steinmetz & Freeman, 2016 [DOI](#); Titley et al., 2007 [DOI](#)). Importantly, changes during this early period significantly shape memory formation and retention, while alterations beyond this timeframe have minimal impact on storage or performance. While our findings align with this framework, extended consolidation periods may involve slower processes, such as systems-level integration into downstream nuclei or cortical regions, contributing to memory stabilization or generalization. Future investigations using chronic in-vivo recordings or live imaging could elucidate these prolonged dynamics and clarify CF transmission's nuanced contributions across phases of cerebellar-dependent memory consolidation.

In conclusion, inhibiting CF transmission during motor learning impairs gain increases specifically during acquisition. These findings provide critical insights into CF signaling's role in cerebellum-dependent learning while underscoring the need for further exploration of how cerebellar circuits contribute to adaptive motor memory over extended timeframes.

Key resources table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Virus	AAV9.CaMKII α .EGFP	Addgene	RRID: Addgene_50469	Plasmid #50469
Virus	AAV1.CaMKII α .eNpHR3.0.EYFP	Addgene	RRID: Addgene_26971	Plasmid #26971
Chemical compound, drug	Zoletil	Virvac		
Chemical compound, drug	Rompun	Bayer		
Chemical compound, drug	Dexamethasone	Samyang Pharmaceutical		
Chemical compound, drug	Meloxicam	Boehringer Ingelheim		
Software, algorithm	ZEN	Zeiss	RRID: SCR_013672	
Software, algorithm	Fiji	http://fiji.sc	RRID: SCR_002285	
Software, algorithm	Patchmaster	HEKA	RRID: SCR_000034	
Software, algorithm	Igor Pro	WaveMetrics	RRID: SCR_000325	
Software, algorithm	GraphPad Prism	GraphPad	RRID: SCR_002798	

Animals and Stereotaxic Surgery for Virus Injection

All experimental procedures were approved by the Seoul National University Institutional Animal Care and Use Committee. First, wild-type mice aged 7–10 weeks were anesthetized using intraperitoneal injections of a Zoletil/Rompun mixture (30 mg / 10 mg/kg). For optogenetic expression in CF, the virus was injected into the IO 2–3 weeks before the behavioral test, as previously described (Kimpo et al., 2014 [DOI](#); Roh et al., 2020 [DOI](#)). Briefly, bilateral injections were made at the midpoint between the edge of the occipital bone and the C1 cervical vertebra. The glass pipette was set at a 55° angle from the vertical and 7° from the midline. After approaching a 3 mm depth, a virus solution containing 100–200 nL of AAV1-CaMkII α -eNpHR3.0-EYFP or AAV9.CaMkII α -EGFP was injected with a Picopump at 5 nL/s. The pipettes were left in place for 10 min before they were removed to minimize backflow.

Stereotaxic Surgery for Optic Cannula Implantation

After the mice were fully recovered (a few days to a week) from the first surgery of virus injection, a second operation of optic cannula implantation (Ø1.25 mm Ceramic Ferrule, L=3.0 mm) near cerebellar flocculus and head fixation was conducted. Unlike the fiber optic cannula, head fixation parts were hand-made. Before the surgery, mice were anesthetized by intraperitoneal injection of Zoletil 50 (Virbac, 15 mg/kg) and xylazine (Rompun, Bayer, 15 mg/kg) mixture. Mice were given 48 hours to recover after surgery.

Confirmation Viral Expression

After completion of all the experiments, sampling using cardiac perfusion is performed. The brain was then extracted and coronal sections were made at 30 µm intervals. Images were acquired and processed using a confocal microscope (Zeiss LSM 7 MP, Carl Zeiss, Jena, Germany) and Zen software (Zeiss). The location of virus expression was confirmed by comparison with the brain atlas (Paxinos and Franklin's The Mouse Brain in Stereotaxic Coordinates 4th Edition by George Paxinos).

Optical suppression of CF terminals in the cerebellar cortex in vivo

To manipulate CF transmission-induced Cs in PCs, AAV1.CaMKIIa.eNpHR3.0.eYFP or AAV8.CaMKIIa.eGFP was bilaterally injected into the IO. A glass micro-pipette was used to inject 200 nL per site. To optically suppress the CF terminals and record CF-PC synaptic response, a small cranial window was made above the cerebellar cortex lobule IV/V a week after the viral injection. After 2 weeks from the viral injection, mice underwent single-unit recordings. The recordings were performed as described in Neural data acquisition and analysis. Cs responses were acquired during 10 seconds of yellow on and off sessions 20 times to test the effect of CF terminal suppression on CF-PC synaptic transmission.

Neural data acquisition and analysis

Single-unit recordings were performed while mice were head-fixed and awake in the recording rack. A Digital Lynx system (Neuralynx, Bozeman, MT) was used to amplify, band pass filter (0.1–8000 Hz for simple spikes, 10–200 Hz for complex spikes), and digitize the electrode recordings. A silicon neural probe (Cambridge NeuroTech) was used to probe and acquire Cs signals. Cs signals were collected at a 32 kHz sampling rate. Cs waveforms were manually selected based on visual inspection of the averaged appearance of characteristic negative Ca^{2+} peaks.

Slice preparation, electrophysiology, and optogenetic validation

An acute brain slice preparation and electrophysiological experiments were carried out as previously described (Lee et al., 2024 [DOI](#)). First, mice aged 9–10 weeks with AAV1.CaMKIIa.eNpHR3.0.WPRE-EYFP injection into their inferior olivary neuron 2–3 weeks before the preparation were anesthetized by isoflurane and briefly decapitated. Then, 250 µm-thick sagittal slices of the cerebellar vermis were obtained from the mice using a vibratome (VT1200S, Leica). The ice-cold cutting solution contained 75 mM sucrose, 75 mM NaCl, 2.5 mM KCl, 7 mM MgCl₂, 0.5 mM CaCl₂, 1.25 mM NaH₂PO₄, 26 mM NaHCO₃, and 25 mM glucose with bubbled 95% O₂ and 5% CO₂. The slices were immediately transferred into the artificial cerebrospinal fluid (ACSF) containing 125 mM NaCl, 2.5 mM KCl, 1 mM MgCl₂, 2 mM CaCl₂, 1.25 mM NaH₂PO₄, 26 mM NaHCO₃, and 10 mM glucose with bubbled 95% O₂ and 5% CO₂. They were allowed to recover at 32 °C for 30 min and at room temperature for 1 hour. All recordings were performed within 8 hr of recovery.

Brain slices were placed in a submerged chamber with perfusion of ACSF for at least 10 min before recording. We used OptoPatcher (HEKA Elektronik) holding recording pipette (2–3 MΩ) filled with internal solution containing 9 mM KCl, 10 mM KOH, 120 mM K-gluconate, 3.48 mM

MgCl₂, 10 mM HEPES, 4 mM NaCl, 4 mM Na₂ATP, 0.4 mM Na₃GTP, and 17.5 mM sucrose (pH 7.25). The stimulation electrode was placed on the granule cell layer near the PC. The stimulation isolator injected a brief current pulse to the stimulation electrode and was controlled using PatchMaster software (HEKA Elektronik). Stimulation intensity (6–30 μ A) was calibrated to minimally evoke CF EPSC. Regarding the laser on trial, a 593.5 nm irradiation from a yellow laser was applied 5 s before CF stimulation. The eEPSC amplitude was averaged over three lasers on and off trials. Under current clamp mode, complex spikes induced by CF stimulation were recorded using the same validation protocol. Overall, the laser illumination (3–5 mW) consistently inhibited CF transmission. Electrophysiological data were acquired using an EPC10 patch-clamp amplifier (HEKA Elektronik) and PatchMaster software (HEKA Elektronik) with a sampling frequency of 20 kHz, and the signals were filtered at 2 kHz. All electrophysiological recordings were acquired from the central cerebellar vermis. The amplitude of EPSC was analyzed using Igor Pro (WaveMetrics).

Optokinetic reflex behavior test

First, two sessions of acclimation were conducted. During each session, the mouse was restrained using a custom-made restrainer for 20 min both with and without light for habituation to the recording environment. Following the procedure reported by (Stahl et al., 2000 [DOI](#)), a calibration was performed. This allowed for the conversion of the dynamics of pupil-to-eye rotation. Next, three basal oculomotor performances including (OKR), vestibulo-ocular reflex in the dark (dVOR), and vestibulo-ocular reflex in the light (lVOR) were measured. During the recording of eye movements, it was necessary to control pupil dilation that was induced by an absence of light in dark conditions. Thus, physostigmine salicylate solution (Eserine; Sigma Millipore) was given to mice under brief isoflurane anesthesia. The concentration of eserine solution was increased from 0.1%, 0.15%, and 0.2% based on the pupil size. Basal oculomotor performances were described in gains. For the actual learning protocol, we adopted OKR and it consisted of five training sessions (10 min per session), six checkup points, and 24 hr of consolidation period. During the training sessions, the mice were visually stimulated with a sinusoidally rotating drum (± 5 degrees). Completely trained mice were placed back into their home cage, which was stored in the dark condition until the last gain checkup points.

Gain Analysis

Gains acquired from basal oculomotor performances and OKR learning were calculated as the ratio of evoked eye movements to the movement of the screen or turn table as visual or vestibular stimuli, respectively. A custom-built LabView (National Instrument) analysis tool was used for all the calculations

Statistical analysis

Graph plotting and statistical analysis were performed using GraphPad Prism (GraphPad Software Inc, CA, USA). The hypothesis was tested by using paired and unpaired t-tests between sample pairs using GraphPad Prism. Statistical significance was set at $p < 0.05$. Asterisks denoted in the graph indicate the statistical significance. * indicates $p < 0.05$, ** < 0.01 , *** < 0.001 , and **** < 0.0001 . The test name and statistical values are presented in each figure legend.

Acknowledgements

This work was supported by National Research Foundation of Korea (NRF) grants funded by the Korean Government (MSIT) (NRF-2018R1A5A2025964 and 2022M3E5E8017970 to Sang Jeong Kim).

References

- Boyden ES, Katoh A, Raymond JL (2004) **Cerebellum-dependent learning: the role of multiple plasticity mechanisms** *Annu Rev Neurosci* **27**:581–609 <https://doi.org/10.1146/annurev.neuro.27.070203.144238>
- Albus J. S. (1971) **A theory of cerebellar function** *Mathematical Biosciences* **10**:25–61 [https://doi.org/10.1016/0025-5564\(71\)90051-4](https://doi.org/10.1016/0025-5564(71)90051-4)
- Attwell P. J. E., Cooke S. F., Yeo C. H. (2002) **Cerebellar function in consolidation of a motor memory** *Neuron* **34**:1011–1020 [https://doi.org/10.1016/S0896-6273\(02\)00719-5](https://doi.org/10.1016/S0896-6273(02)00719-5)
- Bloedel J. R., Bracha V. (1998) **Current concepts of climbing fiber function** *Anatomical Record* **253**:118–126
- Cahill H., Nathans J. (2008) **The Optokinetic Reflex as a Tool for Quantitative Analyses of Nervous System Function in Mice: Application to Genetic and Drug-Induced Variation** *Plos One* **3**:e2055 <https://doi.org/10.1371/journal.pone.0002055>
- Clopath C., Badura A., De Zeeuw C. I., Brunel N. (2014) **A Cerebellar Learning Model of Vestibulo-Ocular Reflex Adaptation in Wild-Type and Mutant Mice** *Journal of Neuroscience* **34**:7203–7215 <https://doi.org/10.1523/Jneurosci.2791-13.2014>
- Coessmans M., Weber J. T., De Zeeuw C. I., Hansel C. (2004) **Bidirectional parallel fiber plasticity in the cerebellum under climbing fiber control** *Neuron* **44**:691–700 <https://doi.org/10.1016/j.neuron.2004.10.031>
- Cooke S. F., Attwell P. J. E., Yeo C. H. (2004) **Temporal properties of cerebellar-dependent memory consolidation** *Journal of Neuroscience* **24**:2934–2941 <https://doi.org/10.1523/Jneurosci.5505-03.2004>
- Eccles J. C., Llinas R., Sasaki K. (1966) **The excitatory synaptic action of climbing fibres on the Purkinje cells of the cerebellum** *J Physiol* **182**:268–296 <https://doi.org/10.1113/jphysiol.1966.sp007824>
- Frens M. A., Mathoera A. L., van der Steen J. (2001) **Floccular complex spike response to transparent retinal slip** *Neuron* **30**:795–801 [https://doi.org/10.1016/S0896-6273\(01\)00321-X](https://doi.org/10.1016/S0896-6273(01)00321-X)
- Goossens H. H. L. M., Hoebeek F. E., van Alphen A. M., van der Steen J., Stahl J. S., De Zeeuw C. I., Frens M. A. (2004) **Simple spike and complex spike activity of floccular Purkinje cells during the optokinetic reflex in mice lacking cerebellar long-term depression (vol 19, pg 687, 2004)** *European Journal of Neuroscience* **19**:1673–1673 <https://doi.org/10.1111/j.1460-9568.2004.03351.x>
- Graf W., Simpson J. I., Leonard C. S. (1988) **Spatial-Organization of Visual Messages of the Rabbits Cerebellar Flocculus. 2. Complex and Simple Spike Responses of Purkinje-Cells** *Journal of Neurophysiology* **60**:2091–2121 <https://doi.org/10.1152/jn.1988.60.6.2091>
- Han V. Z., Zhang Y. P., Bell C. C., Hansel C. (2007) **Synaptic plasticity and calcium signaling in Purkinje cells of the central cerebellar lobes of mormyrid fish** *Journal of Neuroscience*

27:13499–13512 <https://doi.org/10.1523/Jneurosci.2613-07.2007>

Hansel C., Linden D. J. 2000) **Long-term depression of the cerebellar climbing fiber-Purkinje neuron synapse** *Neuron* **26**:473–482 [https://doi.org/10.1016/S0896-6273\(00\)81179-4](https://doi.org/10.1016/S0896-6273(00)81179-4)

Ito M. 1972) **Neural design of the cerebellar motor control system** *Brain Res* **40**:81–84 [https://doi.org/10.1016/0006-8993\(72\)90110-2](https://doi.org/10.1016/0006-8993(72)90110-2)

Jang D. C., Shim H. G., Kim S. J. 2020) **Intrinsic Plasticity of Cerebellar Purkinje Cells Contributes to Motor Memory Consolidation** *Journal of Neuroscience* **40**:4145–4157 <https://doi.org/10.1523/Jneurosci.1651-19.2020>

Ke M. C., Guo C. C., Raymond J. L. 2009) **Elimination of climbing fiber instructive signals during motor learning** *Nature Neuroscience* **12**:1171–U1123 <https://doi.org/10.1038/nn.2366>

Kim O. A., Ohmae S., Medina J. F. 2020) **A cerebello-olivary signal for negative prediction error is sufficient to cause extinction of associative motor learning** *Nature Neuroscience* **23**:1550–U1517 <https://doi.org/10.1038/s41593-020-00732-1>

Kimpo R. R., Rinaldi J. M., Kim C. K., Payne H. L., Raymond J. L. 2014) **Gating of neural error signals during motor learning** *eLife* **3**:e02076 <https://doi.org/10.7554/eLife.02076>

Lee J., Kim S. H., Jang D. C., Jang M., Bak M. S., Shim H. G., Lee Y. S., Kim S. J. 2024) **Intrinsic plasticity of Purkinje cell serves homeostatic regulation of fear memory** *Mol Psychiatry* **29**:247–256 <https://doi.org/10.1038/s41380-023-02320-8>

Marr D. 1969) **A Theory of Cerebellar Cortex** *Journal of Physiology-London* **202**:437 <https://doi.org/10.1113/jphysiol.1969.sp008820>

Medina J. F., Nores W. L., Mauk M. D. 2002) **Inhibition of climbing fibres is a signal for the extinction of conditioned eyelid responses** *Nature* **416**:330–333 <https://doi.org/10.1038/416330a>

Palmer L. M., Clark B. A., Grundemann J., Roth A., Stuart G. J., Hausser M. 2010) **Initiation of simple and complex spikes in cerebellar Purkinje cells** *Journal of Physiology-London* **588**:1709–1717 <https://doi.org/10.1113/jphysiol.2010.188300>

Pham N. C., Kim Y. G., Kim S. J., Kim C. H. 2020) **Differential effects of inferior olive lesion on vestibulo-ocular and optokinetic motor learning** *Neuroreport* **31**:9–16 <https://doi.org/10.1097/Wnr.0000000000001333>

Roh S. E., Kim S. H., Ryu C., Kim C. E., Kim Y. G., Worley P. F., Kim S. K., Kim S. J. 2020) **Direct translation of climbing fiber burst-mediated sensory coding into post-synaptic Purkinje cell dendritic calcium** *eLife* **9**:e61593 <https://doi.org/10.7554/eLife.61593>

Schweigart G., Mergner T., Evdokimidis I., Morand S., Becker W. 1997) **Gaze stabilization by optokinetic reflex (OKR) and vestibulo-ocular reflex (VOR) during active head rotation in man** *Vision Research* **37**:1643–1652 [https://doi.org/10.1016/S0042-6989\(96\)00315-X](https://doi.org/10.1016/S0042-6989(96)00315-X)

Seo J., Kim Y. G., Lee Y.-S., Kim S. J. 2024) **Cerebellar systems consolidation driven by the temporal dynamics of Purkinje cell excitability** *bioRxiv* :2024.2007.2024.604888 <https://doi.org/10.1101/2024.07.24.604888>

- Shaikh A. G., Wong A. L., Optican L. M., Zee D. S. 2017) **Impaired Motor Learning in a Disorder of the Inferior Olive: Is the Cerebellum Confused?** *Cerebellum* **16**:158–167 <https://doi.org/10.1007/s12311-016-0785-x>
- Shutoh F., Ohki M., Kitazawa H., Itohara S., Nagao S. 2006) **Memory trace of motor learning shifts transsynaptically from cerebellar cortex to nuclei for consolidation** *Neuroscience* **139**:767–777 <https://doi.org/10.1016/j.neuroscience.2005.12.035>
- Stahl J. S., van Alphen A. M., De Zeeuw C. I. 2000) **A comparison of video and magnetic search coil recordings of mouse eye movements** *J Neurosci Methods* **99**:101–110 [https://doi.org/10.1016/s0165-0270\(00\)00218-1](https://doi.org/10.1016/s0165-0270(00)00218-1)
- Steinmetz A. B., Freeman J. H. 2016) **Cannabinoid modulation of memory consolidation within the cerebellum** *Neurobiol Learn Mem* **136**:228–235 <https://doi.org/10.1016/j.nlm.2016.11.002>
- Titley H. K., Heskin-Sweezie R., Chung J. Y. J., Kassardjian C. D., Razik F., Broussard D. M. 2007) **Rapid consolidation of motor memory in the vestibuloocular reflex** *Journal of Neurophysiology* **98**:3809–3812 <https://doi.org/10.1152/jn.01056.2007>
- Van Der Giessen R. S., Koekkoek S. K., van Dorp S., De Gruijl J. R., Cupido A., Khosrovani S., Dortland B., Wellershaus K., Degen J., Deuchars J., Fuchs E. C., Monyer H., Willecke K., De Jeu M. T. G., De Zeeuw C. I. 2008) **Role of olivary electrical coupling in cerebellar motor learning** *Neuron* **58**:599–612 <https://doi.org/10.1016/j.neuron.2008.03.016>
- Wada N., Funabiki K., Nakanishi S. 2014) **Role of granule-cell transmission in memory trace of cerebellum-dependent optokinetic motor learning** *Proceedings of the National Academy of Sciences of the United States of America* **111**:5373–5378 <https://doi.org/10.1073/pnas.1402546111>
- Weber J. T., De Zeeuw C. I., Linden D. J., Hansel C. 2003) **Long-term depression of climbing fiber-evoked calcium transients in Purkinje cell dendrites** *Proceedings of the National Academy of Sciences of the United States of America* **100**:2878–2883 <https://doi.org/10.1073/pnas.0536420100>
- Welsh J. P., Harvey J. A. 1998) **Acute inactivation of the inferior olive blocks associative learning** *European Journal of Neuroscience* **10**:3321–3332 <https://doi.org/10.1046/j.1460-9568.1998.00400.x>
- Yang Y., Lisberger S. G. 2014) **Purkinje-cell plasticity and cerebellar motor learning are graded by complex-spike duration** *Nature* **510**:529 <https://doi.org/10.1038/nature13282>
- Zang Y. L., De Schutter E. 2019) **Climbing Fibers Provide Graded Error Signals in Cerebellar Learning** *Frontiers in Systems Neuroscience* **13**:46 <https://doi.org/10.3389/fnsys.2019.00046>

Author information

Jewoo Seo[†]

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine,

Seoul, Republic of Korea
ORCID iD: [0000-0002-9162-848X](https://orcid.org/0000-0002-9162-848X)

[†]Authors contributed equally

Seung Ha Kim[†]

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea
ORCID iD: [0000-0003-0279-8290](https://orcid.org/0000-0003-0279-8290)

[†]Authors contributed equally

Jaegeon Lee[†]

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea
ORCID iD: [0000-0001-7865-3966](https://orcid.org/0000-0001-7865-3966)

[†]Authors contributed equally

Min Seok Kim

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea

Yong-Seok Lee

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea, Memory Network Medical Research Center, Neuroscience Research Institute, Wide River Institute of Immunology, Seoul National University College of Medicine, Seoul, Republic of Korea
ORCID iD: [0000-0002-6217-9574](https://orcid.org/0000-0002-6217-9574)

Sang Jeong Kim

Department of Physiology, Seoul National University College of Medicine, Seoul, Republic of Korea, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Republic of Korea, Memory Network Medical Research Center, Neuroscience Research Institute, Wide River Institute of Immunology, Seoul National University College of Medicine, Seoul, Republic of Korea
ORCID iD: [0000-0001-8931-3713](https://orcid.org/0000-0001-8931-3713)

For correspondence: sangjkim@snu.ac.kr

Editors

Reviewing Editor

Roy Sillitoe

Baylor College of Medicine, Houston, United States of America

Senior Editor

Kate Wassum

University of California, Los Angeles, Los Angeles, United States of America

Reviewer #1 (Public Review):

Summary:

The study by Seo et al highlights knowledge gaps regarding the role of cerebellar complex spike (CS) activity during different phases of learning related to optokinetic reflex (OKR) in mice. The novelty of the approach is twofold: first, specifically perturbing the activity of climbing fibers (CFs) in the flocculus (as opposed to disrupting communication between the inferior olive (IO) and its cerebellar targets globally); and second, examining whether disruption of the CS activity during the putative "consolidation phase" following training affects OKR performance.

The first part of the results provides adequate evidence supporting the notion that optogenetic disruption of normal CF-Purkinje neuron (PN) signaling results in the degradation of OKR performance. As no effects are seen in OKR performance in animals subjected to optogenetic irradiation during the memory consolidation or retrieval phases, the authors conclude that CF function is not essential beyond memory acquisition. However, the manuscript does not provide a sufficiently solid demonstration that their long-term activity manipulation of CF activity is effective, thus undermining the confidence of the conclusions.

Strengths:

The main strength of the work is the aim to examine the specific involvement of the CF activity in the flocculus during distinct phases of learning. This is a challenging goal, due to the technical challenges related to the anatomical location of the flocculus as well as the IO. These obstacles are counterbalanced by the use of a well-established and easy-to-analyse behavioral model (OKR), that can lead to fundamental insights regarding the long-term cerebellar learning process.

Weaknesses:

The impact of the work is diminished by several methodological shortcomings.

Most importantly, the key finding that prolonged optogenetic inhibition of CFs (for 30 min to 6 hours after the training period) must be complemented by the demonstration that the manipulation maintains its efficacy. In its current form, the authors only show inhibition by short-term optogenetic irradiation in the context of electrical-stimulation-evoked CSs in an ex vivo preparation. As the inhibitory effect of even the eNpHR3.0 is greatly diminished during seconds-long stimulations (especially when using the yellow laser as is done in this work (see Zhang, Chuanqiang, et al. "Optimized photo-stimulation of halorhodopsin for long-term neuronal inhibition." *BMC biology* 17.1 (2019): 1-17.), we remain skeptical of the extent of inhibition during the long manipulations. In short, without a demonstration of effective inhibition throughout the putative consolidation phase (for example by showing a significant decrease in CS frequency throughout the irradiation period), the main claim of the manuscript of phase-specific involvement of CF activity in OKR learning can not be considered to be based on evidence.

Second, the choice of viral targeting strategy leaves gaps in the argument for CF-specific mechanisms. CaMKII promoters are not selective for the IO neurons, and even the most precise viral injections always lead to the transfection of neurons in the surrounding brainstem, many of which project to the cerebellar cortex in the form of mossy fibers (MF). Figure 1Bii shows sparsely-labelled CFs in the flocculus, but possibly also MFs. While obtaining homogenous and strong labeling in all floccular CFs might be impossible, at the

very least the authors should demonstrate that their optogenetic manipulation does not affect simple spiking in PNs.

Finally, while the paper explicitly focuses on the effects of CF-evoked complex spikes in the PNs and not, for example, on those mediated by molecular layer interneurons or via direct interaction of the CF with vestibular nuclear neurons, it would be best if these other dimensions of CF involvement in cerebellar learning were candidly discussed.

<https://doi.org/10.7554/eLife.95838.2.sa2>

Reviewer #2 (Public Review):

Summary:

The authors aimed to explore the role of climbing fibers (CFs) in cerebellar learning, with a focus on optokinetic reflex (OKR) adaptation. Their goal was to understand how CF activity influences memory acquisition, memory consolidation, and memory retrieval by optogenetically suppressing CF inputs at various stages of the learning process.

Strengths:

The study addresses a significant question in the cerebellar field by focusing on the specific role of CFs in adaptive learning. The authors use optogenetic tools to manipulate CF activity. This provides a direct method to test the causal relationship between CF activity and learning outcomes.

Weaknesses:

Despite shedding light on the potential role of CFs in cerebellar learning, the study is hampered by significant methodological issues that question the validity of its conclusions. The absence of detailed evidence on the effectiveness of CF suppression and concerns over tissue damage from optogenetic stimulation weakens the argument that CFs are not essential for memory consolidation. These challenges make it difficult to confirm whether the study's objectives were fully met or if the findings conclusively support the authors' claims. The research commendably attempts to unravel the temporal involvement of CFs in learning but also underscores the difficulties in pinpointing specific neural mechanisms that underlie the phases of learning. Addressing these methodological issues, investigating other signals that might instruct consolidation, and understanding CFs' broader impact on various learning behaviors are crucial steps for future studies.

[Editors' note: we have included the original concerns, which the Reviewing Editor agrees with. Methodological concerns remain after revisions.]

<https://doi.org/10.7554/eLife.95838.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

This study presents potentially valuable insights into the role of climbing fibers in cerebellar learning. The main claim is that climbing fiber activity is necessary for optokinetic reflex adaptation, but is dispensable for its long-term consolidation. There is evidence to support the first part of this claim, though it requires a clearer

demonstration of the penetrance and selectivity of the manipulation. However, support for the latter part of the claim is incomplete owing to methodological concerns, including unclear efficacy of longer-duration climbing fiber activity suppression.

We sincerely appreciate the thoughtful feedback provided by the reviewer regarding our study on the role of climbing fibers in cerebellar learning. Each point raised has been carefully considered, and we are committed to addressing them comprehensively. We acknowledge the importance of addressing methodological concerns, particularly regarding the efficacy of long-term suppression of CF activity, as well as ensuring clarity regarding the penetrance and selectivity of our manipulation. To this end, we have outlined plans for substantial revisions to the manuscript to adequately address these issues.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The study by Seo et al highlights knowledge gaps regarding the role of cerebellar complex spike (CS) activity during different phases of learning related to optokinetic reflex (OKR) in mice. The novelty of the approach is twofold: first, specifically perturbing the activity of climbing fibers (CFs) in the flocculus (as opposed to disrupting communication between the inferior olive (IO) and its cerebellar targets globally); and second, examining whether disruption of the CS activity during the putative "consolidation phase" following training affects OKR performance.

The first part of the results provides adequate evidence supporting the notion that optogenetic disruption of normal CF-Purkinje neuron (PN) signaling results in the degradation of OKR performance. As no effects are seen in OKR performance in animals subjected to optogenetic irradiation during the memory consolidation or retrieval phases, the authors conclude that CF function is not essential beyond memory acquisition. However, the manuscript does not provide a sufficiently solid demonstration that their longterm activity manipulation of CF activity is effective, thus undermining the confidence of the conclusions.

Strengths:

The main strength of the work is the aim to examine the specific involvement of the CF activity in the flocculus during distinct phases of learning. This is a challenging goal, due to the technical challenges related to the anatomical location of the flocculus as well as the IO. These obstacles are counterbalanced by the use of a well-established and easy-to-analyse behavioral model (OKR), that can lead to fundamental insights regarding the long-term cerebellar learning process.

Weaknesses:

The impact of the work is diminished by several methodological shortcomings.

*Most importantly, the key finding that prolonged optogenetic inhibition of CFs (for 30 min to 6 hours after the training period) must be complemented by the demonstration that the manipulation maintains its efficacy. In its current form, the authors only show inhibition by short-term optogenetic irradiation in the context of electrical-stimulation-evoked CSs in an ex vivo preparation. As the inhibitory effect of even the eNpHR3.0 is greatly diminished during seconds-long stimulations (especially when using the yellow laser as is done in this work (see Zhang, Chuanqiang, et al. "Optimized photo-stimulation of halorhodopsin for long-term neuronal inhibition." *BMC biology* 17.1 (2019): 1-17.), we remain skeptical of the extent of inhibition during the long manipulations. In short,*

without a demonstration of effective inhibition throughout the putative consolidation phase (for example by showing a significant decrease in CS frequency throughout the irradiation period), the main claim of the manuscript of phase-specific involvement of CF activity in OKR learning cannot be considered to be based on evidence.

Second, the choice of viral targeting strategy leaves gaps in the argument for CF-specific mechanisms. CaMKII promoters are not selective for the IO neurons, and even the most precise viral injections always lead to the transfection of neurons in the surrounding brainstem, many of which project to the cerebellar cortex in the form of mossy fibers (MF). Figure 1Bii shows sparsely-labelled CFs in the flocculus, but possibly also MFs. While obtaining homogenous and strong labeling in all floccular CFs might be impossible, at the very least the authors should demonstrate that their optogenetic manipulation does not affect simple spiking in PNs.

Finally, while the paper explicitly focuses on the effects of CF-evoked complex spikes in the PNs and not, for example, on those mediated by molecular layer interneurons or via direct interaction of the CF with vestibular nuclear neurons, it would be best if these other dimensions of CF involvement in cerebellar learning were candidly discussed.

We appreciate the reviewer's thorough evaluation, which thoughtfully highlights the strengths and areas for improvement in our study.

We agree with the reviewer's recognition of the novelty of our approach, particularly in specifically perturbing climbing fiber (CF) activity in the flocculus and examining its effects across distinct phases of learning. Additionally, our use of the well-established OKR behavior paradigm provides a robust framework for investigating cerebellar learning processes, further strengthening our study.

To address concerns regarding the efficacy of long-term optogenetic inhibition and the specificity of viral targeting, we conducted additional experiments. These include in vivo monitoring of CF activity during the irradiation period, confirming sustained inhibition of complex spikes throughout the consolidation phase. To ensure precise targeting and mitigate potential side effects, such as unintended modification of Purkinje cell (PC) simple spike activity, we demonstrated that optogenetic suppression of CF transmission did not affect simple spike firing. Furthermore, we made additional characterizations to confirm the specificity of viral targeting.

Lastly, we recognize the importance of exploring alternative mechanisms underlying CF involvement in cerebellar learning. Accordingly, we expanded the manuscript to provide a more comprehensive discussion of these mechanisms, offering a clearer perspective on the broader implications of our findings.

Reviewer #2 (Public Review):

Summary:

The authors aimed to explore the role of climbing fibers (CFs) in cerebellar learning, with a focus on optokinetic reflex (OKR) adaptation. Their goal was to understand how CF activity influences memory acquisition, memory consolidation, and memory retrieval by optogenetically suppressing CF inputs at various stages of the learning process.

Strengths:

The study addresses a significant question in the cerebellar field by focusing on the specific role of CFs in adaptive learning. The authors use optogenetic tools to manipulate CF activity. This provides a direct method to test the causal relationship between CF activity and learning outcomes.

Weaknesses:

Despite shedding light on the potential role of CFs in cerebellar learning, the study is hampered by significant methodological issues that question the validity of its conclusions. The absence of detailed evidence on the effectiveness of CF suppression and concerns over tissue damage from optogenetic stimulation weakens the argument that CFs are not essential for memory consolidation. These challenges make it difficult to confirm whether the study's objectives were fully met or if the findings conclusively support the authors' claims. The research commendably attempts to unravel the temporal involvement of CFs in learning but also underscores the difficulties in pinpointing specific neural mechanisms that underlie the phases of learning. Addressing these methodological issues, investigating other signals that might instruct consolidation, and understanding CFs' broader impact on various learning behaviors are crucial steps for future studies.

We appreciate the reviewer's recognition of the significance of our study in addressing the fundamental question of the role of CF in adaptive learning within the cerebellar field. The use of optogenetic tools indeed provides a direct means to investigate the causal relationship between CF activity and learning outcomes.

To address concerns regarding the effectiveness of CF suppression during consolidation, we plan to conduct further in-vivo recordings. These will demonstrate how reliably CF transmission can be suppressed through optogenetic manipulation over an extended period.

In response to the concern about potential tissue damage from laser stimulation, we believe that our optogenetic manipulation was not strong enough to induce significant heat-induced tissue damage in the flocculus. According to Cardin et al. (2010), light applied through an optic fiber may cause critical damage if the intensity exceeds 100 mW, which is eight times stronger than the intensity we used in our OKR experiment. Furthermore, if there had been tissue damage from chronic laser stimulation, we would expect to see impaired long-term memory reflected in abnormal gain retrieval results tested the following day. However, as shown in Figures 2 and 3, there were no significant abnormalities in consolidation percentages even after the optogenetic manipulation.

Finally, we appreciate the reviewer's recognition of the challenges involved in pinpointing specific neural mechanisms. We plan to expand the discussion to address these complexities and outline future research directions.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Inhibitory optogenetic actuators are generally problematic, especially in time frames longer than seconds. If the authors wish to be able to inhibit activity in the flocculus-targeting CFs for a long time, maybe it would make sense to try to retrogradely transfect the IO neurons from the flocculus (using a cre-lox approach) with inhibitory DREADDs. This approach is also full of problems, so the absence or significant decrease in CS activity throughout the period of manipulation must be demonstrated.

In addition to re-examining the strength of the evidence regarding the role of CFs in the consolidation and retrieval phases, the manuscript would benefit from significant reworking of the details in the manuscript and figures. Below is a possibly incomplete list of things we would want to highlight:

(1) While the text states the authors "... verified the potential reduction of Cs firing rate in PCs of awake mice in vivo by inhibiting CF signals", the data nor a figure are shown. This

is of critical importance when judging the reliability of the following results. The data presented in panels Figure 1D-E should also be improved to be more informative, specifically, the waveforms of EPSCs should be shown in higher resolution. We are not informed about how many cells/slices/animals the results are obtained from, nor how many trials were done per condition. Finally, the in vitro data is from vermal Purkinje neurons, while the focus of the work is in the flocculus. Please provide these verifications for the flocculus.

To verify the suppression of complex spike (Cs) activity, we conducted additional in-vivo experiments and added Figure 2, which presents recordings of Cs firing rates from Purkinje cells (PCs) during optogenetic suppression of climbing fiber (CF) activity. These data demonstrate that the suppression specifically and robustly targets Cs activity without affecting simple spike firing, as shown in Figure 2C. The results presented in Figure 2 were acquired at 40 minutes of optostimulation, consistently showing effective suppression of Cs activity throughout this period. While continuous recordings over several hours were not performed, the stability and sustained suppression observed at the 40-minute mark strongly suggest that the manipulation remains effective during the extended durations required for the behavioral tests.

Additionally, we have improved Figure 1D by enhancing the resolution of EPSC waveforms and including more detailed information in the figure legend regarding the number of cells and animals analyzed. For the current-clamp mode data (Figures 1E and F), we clarified the experimental conditions to provide additional context. While the in vitro data were collected from vermal PCs, these experiments were intended to illustrate the fundamental properties of CF-PC transmission.

(2) It is challenging to get a homogenous transfection of all CFs in a given region. To be able to judge the significance of the results, the readers should be provided with material allowing assessing the transfection quality. The images shown in panels Bi-ii are spatially restricted and of too low quality to make judgements. Also, it is not stated whether the images shown are from GFP or NpHR-transfected animals. These different payloads are delivered using different viral capsids (AAV1 vs. AAV9) that have significantly different transfection capacities and results from AAV9-CamKIIIGFP cannot be generalized to AAV1-CamKII-NpHR. Please show the expression for the capsid used with NpHR.

To clarify, the images in Figure Bi-ii are representative of GFP expression in animals transfected using AAV1-CamKII-EGFP. The purpose of these panels is to confirm the successful targeting of the region of interest rather than to evaluate viral tropism or capsid-specific transfection efficiency. Moreover, while the transfection characteristics of AAV1 and AAV9 may differ, the key experimental parameter of effective CF suppression was validated through in-vivo electrophysiological recordings, which robustly confirm the efficacy of NpHR expression.

(3) Finally, please show the location of the optic fiber implant in the flocculus from post-mortem images.

In Figure 3a of our revised manuscript, we added post-mortem histological images showing the exact location of the optic fiber implants in the flocculus. These images provided clear confirmation that the optogenetic stimulation was targeted to the correct anatomical region, ensuring that the observed effects are attributable to CF manipulation in the flocculus.

Reviewer #2 (Recommendations For The Authors):

(1) The efficacy of CF suppression is questionable. The histology in Figure 1 shows that only a handful of CFs are transduced in their approach. This observation casts doubt on

the claimed complete suppression of CF-evoked EPSCs in every recorded PC in the same figure. This necessitates a more detailed explanation for this apparent discrepancy. Also, the absence of current-clamp recordings to measure the effect on CF-evoked complex spiking in PCs and the lack of detail regarding the timing of optogenetic actuation (continuous or pulsed) during these slice experiments are also significant omissions.

We are providing additional in vivo electrophysiological recordings showing sustained CF suppression in awake animals (Figure 2). These recordings will directly demonstrate the extent of CF-evoked complex spike (Cs) suppression.

Moreover, we have included additional data of current-clamp recordings to measure the impact of CF suppression on Cs activity (Figures 1E and 1F). Regarding the timing of the optogenetic actuation, the stimulation was applied continuously in the slice experiments.

(2) The authors claim that their method effectively suppresses CF activity in vivo, yet they do not present any supporting data. Given the histological evidence provided, it's questionable whether their approach truly impacts the CF population broadly, casting doubts on the efficacy of their suppression approach to identify the role of CFs during behavior. To address these concerns, further experiments and detailed quantification are essential to validate the extent and uniformity of CF suppression achieved.

As we responded earlier, we conducted additional in-vivo experiments with continuous recordings of CF-evoked complex spike (Cs) activity during optogenetic suppression (Figure 2). These data directly demonstrate effective and sustained inhibition of CF transmission throughout the behavioral experiments. Quantification of CF suppression revealed consistent inhibition across the manipulation period, with no observable alterations in Purkinje cell simple spike firing rates, confirming that our intervention specifically targeted CF activity without off-target effects. In addition to the in-vivo data, the in-vitro data presented in Figure 1 (lines 107~116) further validate the efficacy of our optogenetic manipulation, showing consistent suppression of CF transmission without any failures. These findings collectively confirm the reliability and specificity of our suppression approach for studying CF contributions to behavior.

(3) To optogenetically test the role of CFs in memory consolidation, the authors deliver continuous, high-power light to the flocculus (13 mW for 6 hrs). This extends well beyond typical experimental conditions. The sustained nature of the light exposure thus brings into question the consistency and reliability of CF suppression over time. Firstly, it is imperative to determine whether CF activity is suppressed throughout this extended period. Secondly, the intensity and duration of light exposure carry a significant risk of causing extensive damage to the surrounding tissue. Given these concerns, a thorough histological examination is warranted to assess the potential adverse effects on tissue integrity. Such an analysis is crucial not only for validating the experimental outcomes but also for ensuring that the observed effects are not confounded by light-induced tissue damage.

To address whether CF activity is suppressed throughout the extended period, we included new in-vivo recordings demonstrating robust suppression of CF transmission, as evidenced by inhibited complex spikes sustained at 40 minutes of optostimulation. Regarding potential tissue damage, our optogenetic protocol used a light intensity (13 mW), which is much lower than the 75 mW threshold reported by Cardin et al. (2010) as sufficient to maintain normal neuronal activity. Moreover, critical damage typically requires intensities exceeding 100 mW for several hours (Cardin, Jessica A., et al. "Targeted optogenetic stimulation and recording of neurons in vivo using cell-type-specific expression of Channelrhodopsin-2." *Nature protocols* 5.2 (2010): 247-254.). Finally, we observed no abnormalities in long-term memory

consolidation or gain retrieval (Figures 3C, 4C, 4F), further supporting that our light stimulation did not induce tissue damage.

(4) The generalizability of their findings to various learning behaviors remains uncertain. Given that the flocculus plays a role in vestibulo-ocular reflex (VOR) adaptation, which encompasses both CF-dependent and CF-independent learning types (gain increase and gain decrease, respectively), this system could offer a more feasible approach for investigating hypotheses about the role of CFs in guiding distinct learning processes.

In response to the reviewer's comment on the generalizability of our findings to learning behaviors involving both CF-dependent and CF-independent mechanisms, we acknowledge the importance of examining these dynamics in cerebellar motor adaptation systems, such as the OKR. Although our study used an OKR task, findings from VOR studies apply here. Ke et al. (2009) demonstrated that VOR gain increases (CF-dependent) and gain decreases (CF-independent) involve distinct plasticity processes (Ke, Michael C., Cong C. Guo, and Jennifer L. Raymond. "Elimination of climbing fiber instructive signals during motor learning." *Nature neuroscience* 12.9 (2009): 1171-1179), suggesting that CF engagement is task-dependent, particularly for larger error signals that require CF-guided adaptation.

Similarly, our OKR findings suggest that CF-dependent pathways are likely used for large, persistent errors, whereas CF-independent mechanisms may drive more gradual adjustments. This alignment between OKR and VOR systems supports the generalizability of CF-selective adaptation across cerebellar learning tasks. We have elaborated on this point in our revised manuscript (lines 219~237), clarifying how CF-dependent and CF-independent mechanisms can generalize across motor learning contexts in the cerebellum.

(5) The acute effect of CF suppression on OKR eye movements warrants investigation. If OKR eye movements are altered by their method, this could complicate the interpretation of their results.

During our experiments, we monitored ocular movements during CF optogenetic manipulation and found no aberrant effects, such as nystagmus. As shown in Figures 4G and 4H, disrupting CF signaling during gain retrieval did not alter the gain, confirming that our manipulation neither acutely affects ocular reflexes nor induces abnormal eye movement. Therefore, it leads to the conclusion that the observed effects are specific to learning and memory processes.

(6) The authors raise the potential issue of inducing presynaptic LTD in CFs. Can they be sure that their manipulation doesn't generate a similar effect? Additional controls or techniques to accurately interpret the results are needed considering this concern.

However, our discussion does not claim that optogenetic suppression directly induces CF-LTD. Instead, we posit that CF suppression may have mimicked the functional consequences of CFLTD, such as reduced complex spike (Cs) activity and associated calcium signaling. This, in turn, may have indirectly interfered with the induction of parallel fiber-Purkinje cell (PF-PC) LTD, thereby preventing gain enhancement during learning.

This hypothesis is consistent with previous studies highlighting the interplay between CF and PF synaptic plasticity in cerebellar motor learning. For example, Hansel and Linden (2000) and Weber et al. (2003) discuss how changes at CF synapses can modulate Cs waveforms and calcium dynamics, which are critical for PF-PC LTD. Coesmans et al. (2004) and Han et al. (2007) further elaborate on the necessity of CF input for effective PF-PC LTD induction during learning tasks such as retinal slip correction.

While our experiments were not designed to directly measure CF-LTD, the observed prevention of gain enhancement aligns with the hypothesis that CF suppression functionally disrupted downstream PF-PC LTD. We have clarified these points in our revised manuscript (lines 250~258) to avoid misunderstanding.

(7) The specific timeframe for OKR consolidation remains uncertain, with evidence from numerous studies indicating that cerebellar memory consolidation unfolds over several days. Therefore, a more thorough investigation into these extended durations, supported by control experiments to validate the outcomes, would significantly strengthen the study's conclusions, and provide clearer insights into the consolidation process of OKR learning.

Our current study specifically focused on the early phase of the post-learning period, as supported by findings from several studies: Cooke et al., (2004); Titley et al., (2007); Steinmetz et al., (2016); Seo et al., (2024)

These studies collectively indicate that cerebellar-dependent memory consolidation—including OKR—can occur rapidly during the early consolidation phase. While the specific mechanisms examined in these studies vary (e.g., synaptic plasticity, intrinsic plasticity, or circuit-level changes), they consistently demonstrate that modifications in the cerebellum after the early consolidation period no longer influence memory storage or performance. This evidence strongly supports the relevance of our experimental focus and the timing of our interventions.

We acknowledge the importance of investigating extended consolidation periods, which could indeed provide additional insights. However, given our current aims, the rapid consolidation dynamics observed in the early phase are most relevant to the questions addressed in this study. We have elaborated on these matter in our revised manuscript (lines 273~283).

(8) Issues around whether the authors have control over CF activity with their optogenetic intervention raise questions of whether learning can be recovered during the training procedure if the optogenetic stimuli are halted. Specifically, if suppression is applied for three blocks (what the authors refer to as "sessions") during the training procedure and then ceases, does learning rapidly recover in the immediately following blocks?

While we did not directly examine the restoration of learning capability within the same training session following the cessation of optogenetic inhibition, we believe several aspects of our experimental design and insights from prior studies support our interpretation.

Our optogenetic intervention specifically targeted Purkinje cells (PCs) in the flocculus and was applied continuously during designated training sessions to modulate cerebellar activity. Notably, Medina et al. (2001) demonstrated that transient inactivation of the cerebellar cortex impairs the expression of learned responses but does not disrupt the underlying plasticity mechanisms (Medina, Javier F., Keith S. Garcia, and Michael D. Mauk. "A mechanism for savings in the cerebellum." *Journal of Neuroscience* 21.11 (2001): 4081-4089.). This finding suggests that cerebellar plasticity remains intact and functional even after transient perturbations.

Therefore, it is plausible that once optogenetic inhibition is lifted, the cerebellar network regains its capacity for learning and adaptation, as the intrinsic plasticity and memory encoding processes remain preserved. While we acknowledge that direct experimental confirmation of rapid recovery in our setup was not performed, this interpretation is consistent with our experimental framework and the broader literature.

(9) The study does not fully explore the instructive signals/mechanisms underlying the memory consolidation process. A detailed investigation into potential instructive signals for consolidation beyond CF-induced signaling, like the simple spiking of PCs, could significantly enhance the study's conclusions. Indeed, there is currently no evidence to suggest that CFs play a role in the consolidation phase anyway so testing their role seems a bit of a strawman argument.

While our study primarily focused on characterizing CF-dependent pathways, we acknowledge that memory consolidation is likely driven by a multifaceted interplay of instructive signals beyond CF-induced mechanisms. In particular, Purkinje cell (PC) simple spiking may act as a critical signal during the consolidation phase, either complementing or functioning independently of CF input. Emerging evidence suggests that simple spiking can modulate downstream circuitry in ways that stabilize and strengthen memory traces.

To address this, we have expanded the discussion in the revised manuscript to explore potential instructive signals for consolidation, including PC simple spiking, local circuit plasticity within the cerebellar cortex, and its interaction with the cerebellar nuclei. We propose that these mechanisms collectively contribute to the transfer and stabilization of motor memory, offering a more comprehensive framework for understanding consolidation. We have elaborated on these matter in our revised manuscript (lines 238~250).

(10) Previous reports have highlighted the necessity of CF activity for extinction/memory maintenance (Medina et al. 2002; Kim et al. 2020). That is, the absence of CF activity is consequential for cerebellar function. These results present a potential contrast to the findings reported in this current study. This discrepancy raises important questions about the experimental conditions, methodologies, and interpretations of CF function across different studies. A thorough discussion comparing these divergent outcomes is essential, as it could elucidate the specific contexts or conditions under which CF activity influences memory processes.

We acknowledge that previous studies (Medina et al., 2002; Kim et al., 2020) have suggested a role for climbing fiber (CF) activity in extinction. However, our study specifically focuses on the acquisition phase of motor learning and does not extend to extinction or maintenance. As such, we have revised our discussion to limit interpretations strictly to the scope of our findings and removed references to extinction.

The discrepancies between our results and prior work may arise from differences in methodologies and behavioral paradigms. For instance, we utilized optogenetic inhibition to achieve precise temporal and spatial control of CF activity, whereas previous studies employed pharmacological or lesion methods that may have broader effects on the cerebellar circuitry. Additionally, differences in behavioral paradigms, such as the optokinetic reflex (OKR) task used in our study compared to the eye-blink conditioning tasks in prior studies, may demand distinct roles for CF signaling depending on the specific requirements for error correction and adaptation.

This clarification is now incorporated into our revised manuscript, and the discussion has been streamlined to focus on the phase-specific role of CF activity during acquisition without extending to extinction or maintenance (lines 259~270).

<https://doi.org/10.7554/eLife.95838.2.sa0>